HW #8

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. Let A be an integral domain, and let $|\cdot| : A \to \mathbb{R}_{\geq 0}$ be a non-archimedean absolute value.

(1) Define
$$A^\circ := \{a \in A \ : \ |a| \leq 1\},$$
$$A^{\circ\circ} := \{a \in A \ : \ |a| < 1\}.$$
Show that A° is a subring of A, and $A^{\circ\circ} \subset A^\circ$ is a prime ideal.

(2) On $A[X]$, define the **Gauss norm** $|\cdot|_{\text{Gauss}} : A[X] \to \mathbb{R}_{\geq 0}$ as
$$\left|a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0\right|_{\text{Gauss}} := \max(|a_n|, \ |a_{n-1}|, \ \cdots, \ |a_0|).$$
Show that the Gauss norm is a non-archimedean absolute value on $A[X]$.

Hint. By taking $\mathrm{Frac}(A)$, we can assume that A is a field. The only nontrivial axiom is $|f(X)g(X)|_{\text{Gauss}} = |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$ for $f(X), g(X) \in A[X]$. It is easy to show $|f(X)g(X)|_{\text{Gauss}} \leq |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$. If $|f(X)g(X)|_{\text{Gauss}} < |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$ for some $f(X), g(X) \in A[X]$, after multiplying by an appropriate element in A, one can assume, without loss of generality, that $|f(X)|_{\text{Gauss}} = |g(X)|_{\text{Gauss}} = 1$. This implies that
$$f(X), g(X) \in A^\circ[X], \ f(X)g(X) \in A^{\circ\circ}[X]$$
Deduce a contradiction by showing that $A^\circ[X]/A^{\circ\circ}[X] \cong (A^\circ/A^{\circ\circ})[X]$ and using (1).

(3) Use (2) to show the following[1].

> **Gauss's lemma**. Let A be a UFD. A polynomial $f(X) \in A[X]$ is called **primitive** if the gcd of its coefficients is 1. Then, the product of primitive polynomials is primitive.

Hint. For an irreducible element $\pi \in A$, there is a normalized discrete valuation v_π on A. Taking $0 < \epsilon < 1$, one can define a non-archimedean absolute value $|\cdot|_\pi$ on A by $|x|_\pi := \epsilon^{v_\pi(x)}$.

Proof.

(1) It is clear that A° is additively and multiplicatively closed. It is also clear that, if $xy \in A^{\circ\circ}$ for $x, y \in A^\circ$, it means that $|x||y| < 1$, so either $|x|$ or $|y|$ has to be strictly less than 1, so $A^{\circ\circ} \subset A^\circ$ is a prime ideal (that it is an ideal is also clear).

(2) As noted in Hint, it suffices to show that the same formula defines a non-archimedean absolute value on $\mathrm{Frac}(A)[X]$, which contains $A[X]$ as a subring. Therefore, let's assume that A is a field. Let's check the three axioms.
- For $f(X) = a_n X^n + \cdots + a_0 \in A[X]$, if $|f(X)|_{\text{Gauss}} = 0$, then $|a_n| = \cdots = |a_0| = 0$, so $a_n = \cdots = a_0 = 0$, so $f(X) = 0$; the converse is easy.

[1] A special version was stated in Theorem 1.11.

- For $f(X) = a_nX^n + \cdots + a_0$ and $g(X) = b_nX^n + \cdots + b_0$ (leading terms may be zero; we can use the same formula to compute the Gauss norm even after adding some zero terms),

$$|f(X) + g(X)|_{\text{Gauss}} = \max(|a_n + b_n|, \cdots, |a_0 + b_0|)$$
$$\leq \max(|a_n|, |b_n|, \cdots, |a_0|, |b_0|) = \max(|f(X)|_{\text{Gauss}}, |g(X)|_{\text{Gauss}}).$$

- For $f(X) = a_nX^n + \cdots + a_0$ and $g(X) = b_mX^m + \cdots + b_0$, $f(X)g(X) = c_{m+n}X^{m+n} + \cdots + c_0$, where

$$c_i = \sum_{j=0}^{i} a_j b_{i-j}$$

where $a_j = 0$ for $j > n$ and $b_j = 0$ for $j > m$. Thus

$$|c_i| \leq \max\{|a_j||b_{i-j}| \ : \ 0 \leq j \leq i\}.$$

Thus

$$|f(X)g(X)|_{\text{Gauss}} = \max(|c_{m+n}|, \cdots, |c_0|) \leq \max\{|a_j||b_k| \ : \ 0 \leq j \leq n, \ 0 \leq k \leq m\}$$
$$= \max(|a_n|, \cdots, |a_0|) \max(|b_m|, \cdots, |b_0|) = |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}.$$

Suppose that the equality does not hold for some $f(X)$ and $g(X)$, i.e., $|f(X)g(X)|_{\text{Gauss}} < |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$. Necessarily, $|f(X)|_{\text{Gauss}}, |g(X)|_{\text{Gauss}} \neq 0$. Thus, there exist $a, b \in A$ such that $|af(X)|_{\text{Gauss}} = |bg(X)|_{\text{Gauss}} = 1$ (e.g., $a = a_i^{-1}$ for $0 \leq i \leq n$ such that $|a_i| = \max(|a_n|, \cdots, |a_0|)$, and similarly for b). As $|abf(X)g(X)|_{\text{Gauss}} = |a||b||f(X)g(X)|_{\text{Gauss}} < |a||b||f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}} = 1$, we may assume that $|f(X)|_{\text{Gauss}} = |g(X)|_{\text{Gauss}} = 1$. This implies that $f(X), g(X) \in A^\circ[X]$, and also that $f(X)g(X) \in A^{\circ\circ}[X]$. Therefore, we will obtain a contradiction if we show that $A^{\circ\circ}[X] \subset A^\circ[X]$ is a prime ideal. By the inequality $|f(X)g(X)|_{\text{Gauss}} \leq |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$ we showed above, it is clear that $A^{\circ\circ}[X]$ is an ideal of $A^\circ[X]$. Moreover, as $A^{\circ\circ} \subset A^\circ$ is a prime ideal by (1), $A^\circ/A^{\circ\circ} =: R$ is an integral domain. Consider the quotient map $A^\circ[X] \to R[X]$. It is surjective, and its kernel is clearly $A^{\circ\circ}[X]$ (this is another way of showing that $A^{\circ\circ}[X]$ is an ideal). As $R[X]$ is also an integral domain, it follows that $A^{\circ\circ}[X]$ is a prime ideal.

(3) Let $f(X), g(X) \in A[X]$ be primitive polynomials. We want to show that $f(X)g(X)$ is primitive. This is equivalent to showing that the gcd of the coefficients of $f(X)g(X)$ is not divisible by any irreducible element of A. For an irreducible element $\pi \in A$, there is a normalized discrete valuation $v_\pi : A \to \mathbb{Z}_{\geq 0} \cup \{\infty\}$ given by, for $x \neq 0$, $\pi^{v_\pi(x)}|x$ and $\pi^{v_\pi(x)+1} \nmid x$. Then $|\cdot|_\pi$ as in Hint defines a non-archimedean absolute value on A. It suffices to show that, if $|\cdot|_{\text{Gauss}}$ is the corresponding Gauss norm on $A[X]$, that $|f(X)g(X)|_{\text{Gauss}} = 1$. However, $|f(X)|_{\text{Gauss}} = |g(X)|_{\text{Gauss}} = 1$ by primitivity, so $|f(X)g(X)|_{\text{Gauss}} = 1$ by (2).

□

Question 2.

(1) Let L/K be a Galois extension of p-adic local fields, and let $|\cdot|$ is a non-archimedean absolute value on L. Show that any element $\sigma \in \text{Gal}(L/K)$ is an isometry. In other words, show that $|x - y| = |\sigma(x) - \sigma(y)|$ for any $x, y \in L$.

Hint. Use Theorem 4.44.

(2) **(Krasner's lemma)** Let K be a p-adic local field. Let $f(X) \in K[X]$ be an irreducible separable monic polynomial, and let L/K be a separable extension of local fields containing the splitting field of $f(X)$; in other words, $f(X)$ factorizes in $L[X]$ as

$$f(X) = \prod_{i=1}^{n}(X - \alpha_i), \quad \alpha_i \in L.$$

Let $|\cdot|$ be a non-archimedean absolute value on L. Let $\beta \in L$ be such that

$$|\beta - \alpha_1| < |\alpha_j - \alpha_1| \quad \text{for all } j \neq 1.$$

Show that $K(\alpha_1) \subset K(\beta)$.

Hint. Enlarge L so that L/K is Galois. If $K(\alpha_1) \not\subset K(\beta)$, there should be $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(\alpha_1) \neq \alpha_1$ and $\sigma(\beta) = \beta$. Use (1) to deduce a contradiction.

(3) Let K be a p-adic local field, and $|\cdot|$ be a non-archimedean absolute value on K (or its unique extension on any finite extension of K). Let L/K be a finite Galois extension. Show that there exists $0 < c_{L/K} \leq 1$ (depending only on L, K, $|\cdot|$) such that the following holds.

> **Lemma** (Fontaine). Let E/K be any Galois extension, and let $0 < \epsilon < c_{L/K}$. Define
> $$\mathfrak{m}_E^\epsilon = \{x \in \mathcal{O}_E \ : \ |x| \leq \epsilon\}.$$
> If there exists an $\mathcal{O}_K$-algebra homomorphism $\mathcal{O}_L \to \mathcal{O}_E/\mathfrak{m}_E^\epsilon$, there exists an $\mathcal{O}_K$-algebra homomorphism $\mathcal{O}_L \to \mathcal{O}_E$.

Hint. By Theorem 4.38(1), $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ for some $\alpha \in \mathcal{O}_L$. Let $f(X) \in \mathcal{O}_K[X]$ be the minimal polynomial of α over $\mathcal{O}_K$. What the problem is asking is, if there is $\beta \in \mathcal{O}_E$ such that $|f(\beta)| \leq \epsilon$, whether there is $\beta' \in \mathcal{O}_E$ such that $f(\beta') = 0$. Use (2).

Proof.

(1) It suffices to show that $|x| = |\sigma(x)|$ for $x \in L$. This means that for any irreducible separable polynomial $f(X) \in K[X]$, all roots have the same absolute values. By Theorem 4.44(4), $f(X)$ has only one slope, so by Theorem 4.44(1), all roots have the same valuation, so the same absolute values.

(2) One can always make L bigger, so assume that L/K is Galois. If $K(\alpha_1) \not\subset K(\beta)$, then there exists $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(\alpha_1) \neq \alpha_1$ and $\sigma(\beta) = \beta$. In this case, $\sigma(\alpha_1) = \alpha_j$ for some $j \neq 1$. Therefore, by (1), $|\beta - \alpha_1| = |\sigma(\beta) - \sigma(\alpha_1)| = |\beta - \alpha_j|$. However, as $|\beta - \alpha_1| < |\alpha_j - \alpha_1|$, $|\beta - \alpha_j| = |(\beta - \alpha_1) - (\alpha_j - \alpha_1)| = |\alpha_j - \alpha_1| > |\beta - \alpha_1|$, a contradiction.

Here, we are using the fact that, for a non-archimedean absolute value, if $|x| > |y|$, then $|x + y| = |x|$; this is because $|x + y| \leq \max(|x|, |y|) = |x|$ by the strong triangle inequality, and $|x| \leq \max(|x + y|, |-y|) = \max(|x + y|, |y|)$, and as $|x| > |y|$, the RHS must be $|x + y|$, so that $|x| \leq |x + y|$.

(3) Firstly, as per Hint, we know that there is $\beta \in \mathcal{O}_E$ such that $|f(\beta)| \leq \epsilon$. Let us take a large enough field F that contains both E and L as subfields. Then, $f(X) = \prod_{i=1}^{n}(X - \alpha_i)$ for $\alpha_i \in F$, with $\alpha_1 = \alpha$. Then $|\prod_{i=1}^{n}(\beta - \alpha_i)| \leq \epsilon$. Therefore, by pigeonhole principle, there exists some $1 \leq j \leq n$ such that $|\beta - \alpha_j| \leq \epsilon^{1/n}$.

We can now let

$$c_{L/K} = \min\{|\alpha_i - \alpha_j| \ : \ 1 \leq i < j \leq n\}^n.$$

Note first that this is ≤ 1, as each $|\alpha_i - \alpha_j| \leq 1$. Now for any $I \neq j$,

$$|\beta - \alpha_j| < c_{L/K}^{1/n} \leq |\alpha_i - \alpha_j|.$$

By (2), it follows that $K(\alpha_j) \subset K(\beta) \subset E \subset F$. Thus $\alpha_j \in E$, and as α_j is integral over $\mathcal{O}_K$, $\alpha_j \in \mathcal{O}_E$ by Theorem 4.35. Take $\beta' = \alpha_j$ as in Hint, we are done.

$\square$